

7091CEM Lab Session 3b

Machine Learning with Weka

In this lab you will carry out all the Machine Learning Tasks below.

If you have any additional questions on this lab email me on ab0487@coventry.ac.uk.

Add your solution on the AULA feed using the tag **#lab3b**.

Basic Tasks – ML

Task a)



Give six fields that big data machine learning can be used in? You should also give a brief description of at least two examples of how machine learning can be used for in these areas.

Answer
Medicine
Diagnostic
Logistics

Military
Weapons
Planning

Task b)



An insurance company has the data in the form below and wishes to use an intelligent approach to determine what policies people will buy.

Age	Income	Gender	Previous Claims	Policy
22	20,000	Female	0	Car
35	25,000	Female	2	Pet
65	15,000	Male	0	None
34	80,000	Male	3	Car and Holiday

Answer

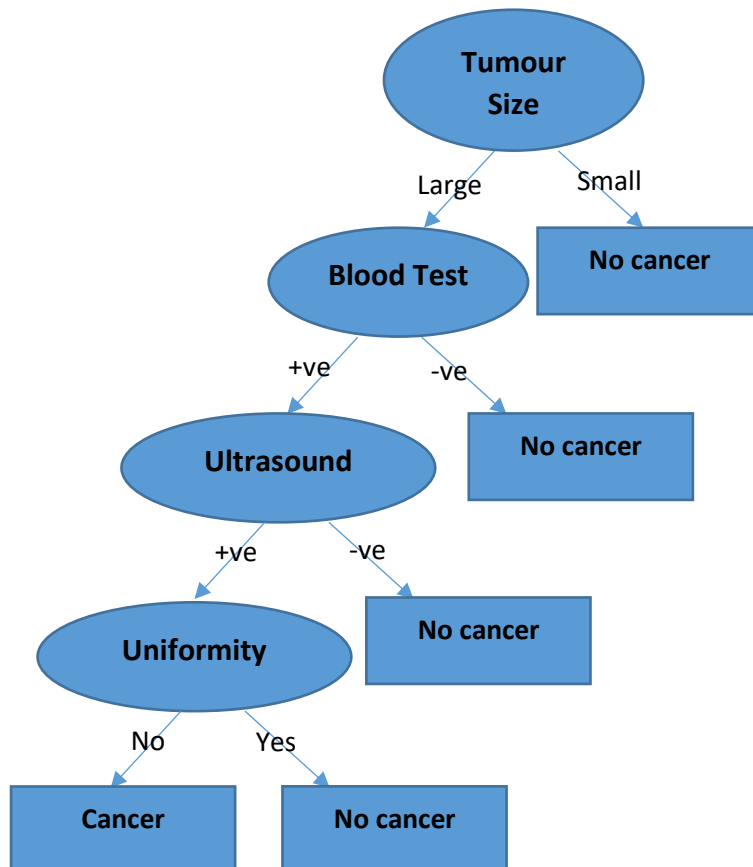
Use a supervised approach to classify the insurance policies.

Task c)



Create a simple decision tree for a cancer diagnostic system using the following findings from a clinical study.

Tumour Size	Blood test	Poor diet	Ultrasound test	Uniformity of Cell thickness	Diagnosis
Large	Positive	Yes/No	Positive	No	Cancer
Small	Negative	Yes/No	negative	Yes	Non-Cancer



Advanced Tasks – Weka

Running WEKA

Get into Apps Anywhere from start button
Search for WEKA.
Install WEKA.

You can also download and install WEKA from its website
<https://www.cs.waikato.ac.nz/ml/weka/index.html>.

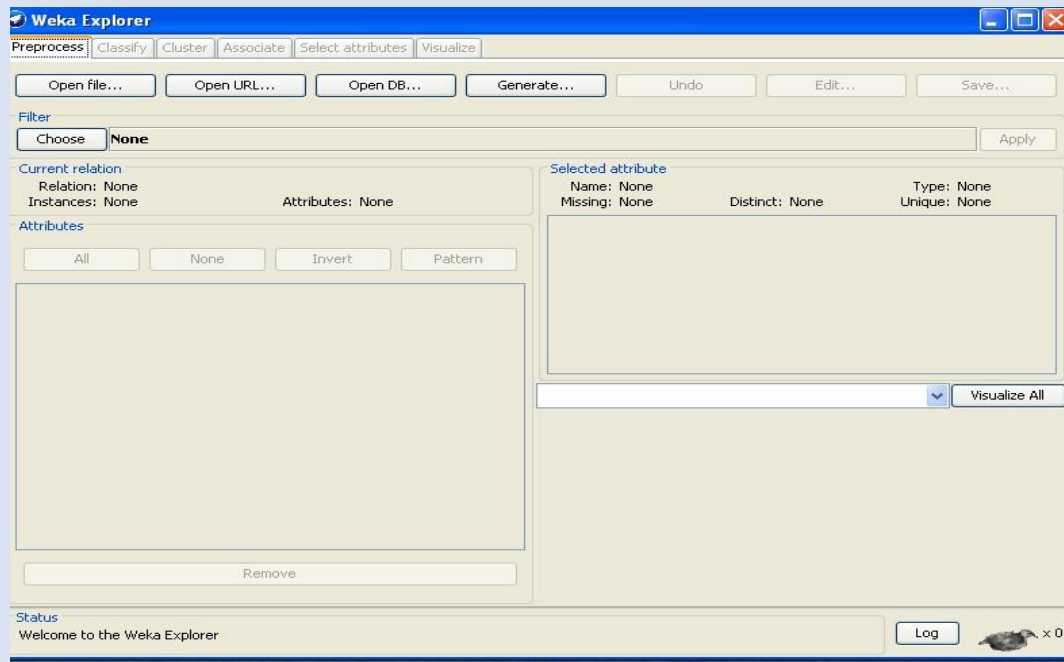
Then the first interface window appears: Weka **GUI Chooser**.



Download the data

From AULA download, unzip and extract the iris.arff and credit.arff data.
This iris data is features for different types of iris and the credit data includes features to determine if a person is credit worthy.

We will explore how we can preprocess data in the future sessions.



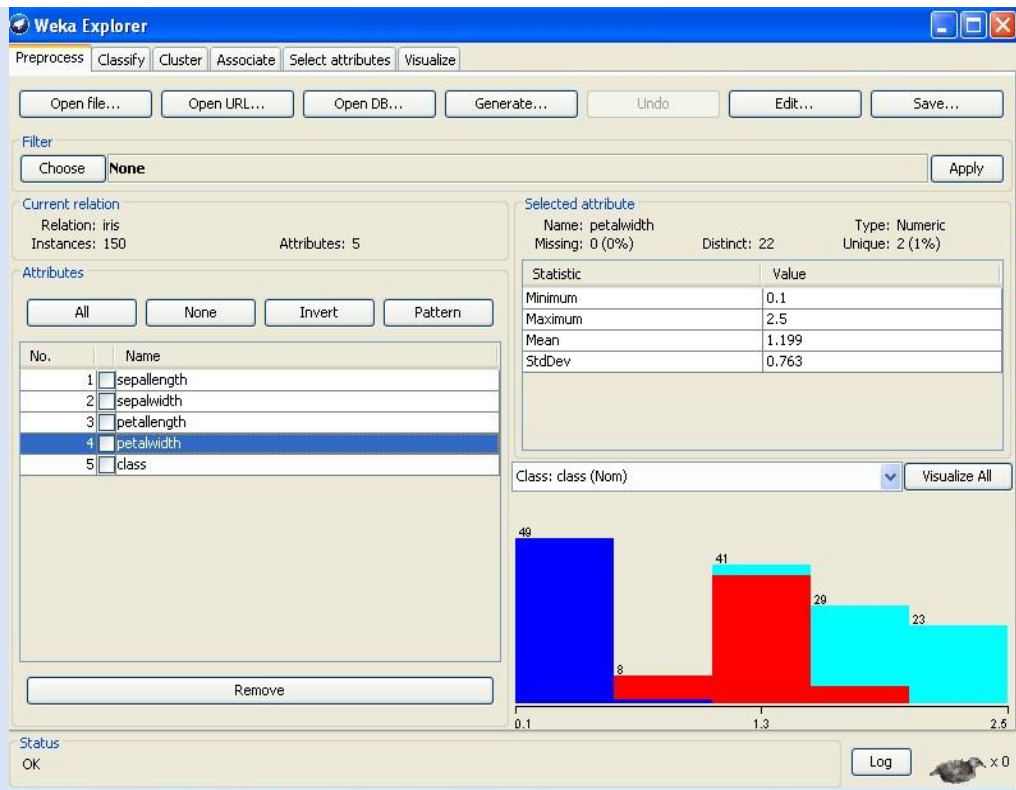
Press the open file button

Go to the directory containing the iris.arff file

Weka supports various formats including .arff and .csv. These are simply data files.

Click on iris.arff

The iris problem is famous in machine learning. It is typically used when someone invents a new algorithm as it is a good benchmark data. It uses 4 attributes to classify what type of iris the flower is.



Click on the different attributes.

The different colours are the different classes of iris. You should see that it is not easy to split the data into the three iris classes.

Activity

What are the three different types of iris?

Answer:

No.	Label	Count
1	Iris-setosa	50
2	Iris-versicolor	50
3	Iris-virginica	50

sepalwidth
sepalwidth
petalwidth
petalwidth

In the selected attribute panel there is information on the data. This is valuable as it can give you if you have any extremely high values that might prevent your machine learning approach from learning the data.

Activity

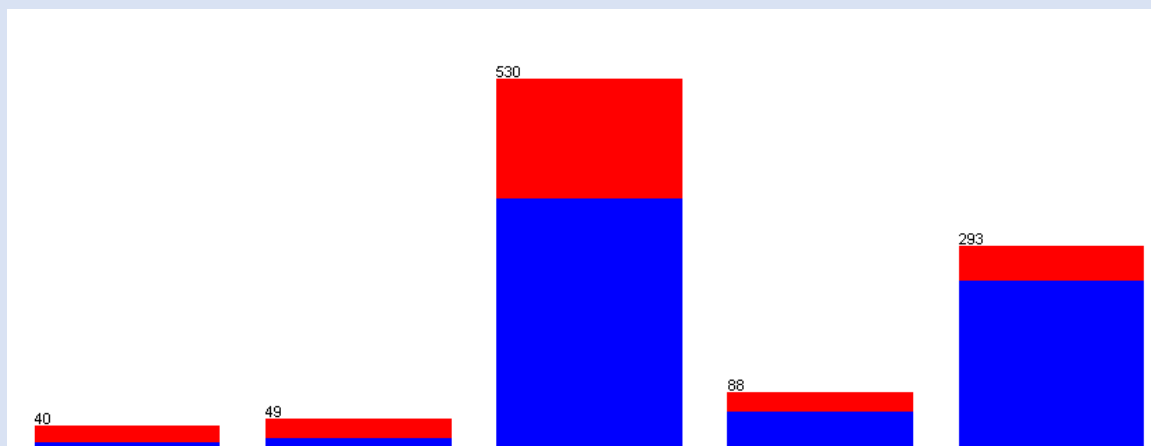
Now open the credit.arff.

What are some of the attributes used to determine credit worthiness and what are the classes?

Produce a screenshot of the bar-chart of the values for the credit_history attribute.

Answer

No.		Name
1	☐	checking_status
2	☐	duration
3	☐	credit_history
4	☐	purpose
5	☐	credit_amount
6	☐	savings_status
7	☐	employment
8	☐	installment_commitment
9	☐	personal_status
10	☐	other_parties
11	☐	residence_since
12	☐	property_magnitude
13	☐	age
14	☐	other_payment_plans
15	☐	housing
16	☐	existing_credits
17	☐	job
18	☐	num_dependents
19	☐	own_telephone
20	☐	foreign_worker



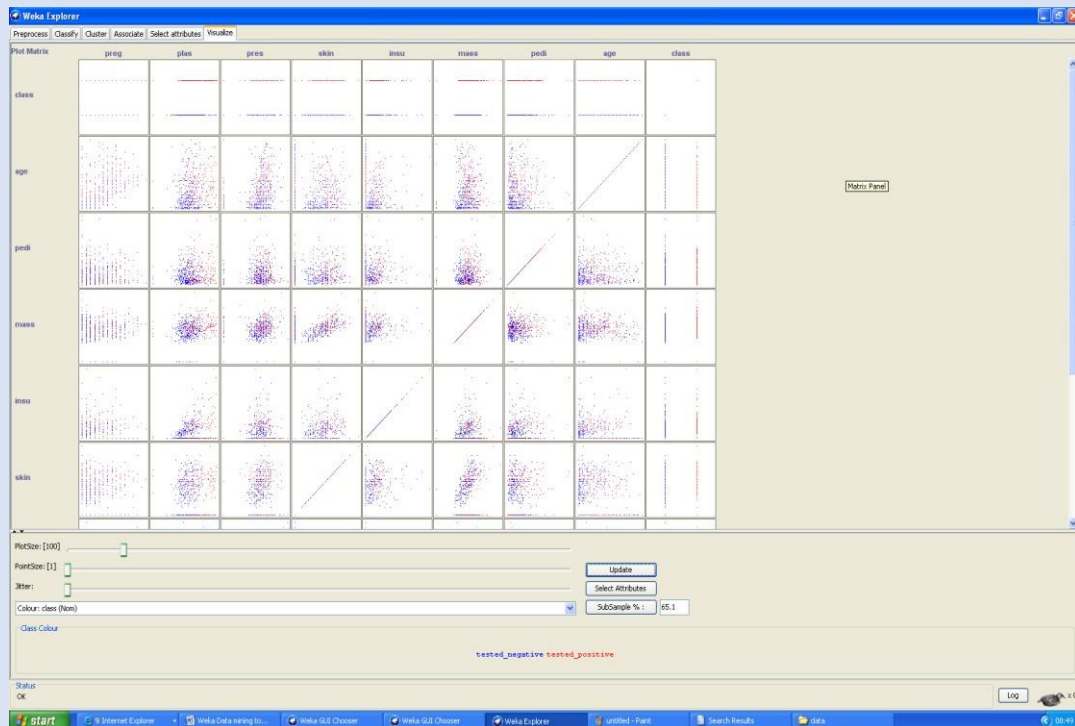
Task b)



WEKA Visualization

For the iris data click on the visualize tab.

This plots the attribute against attribute. You can see if there is any relationship between these attributes.

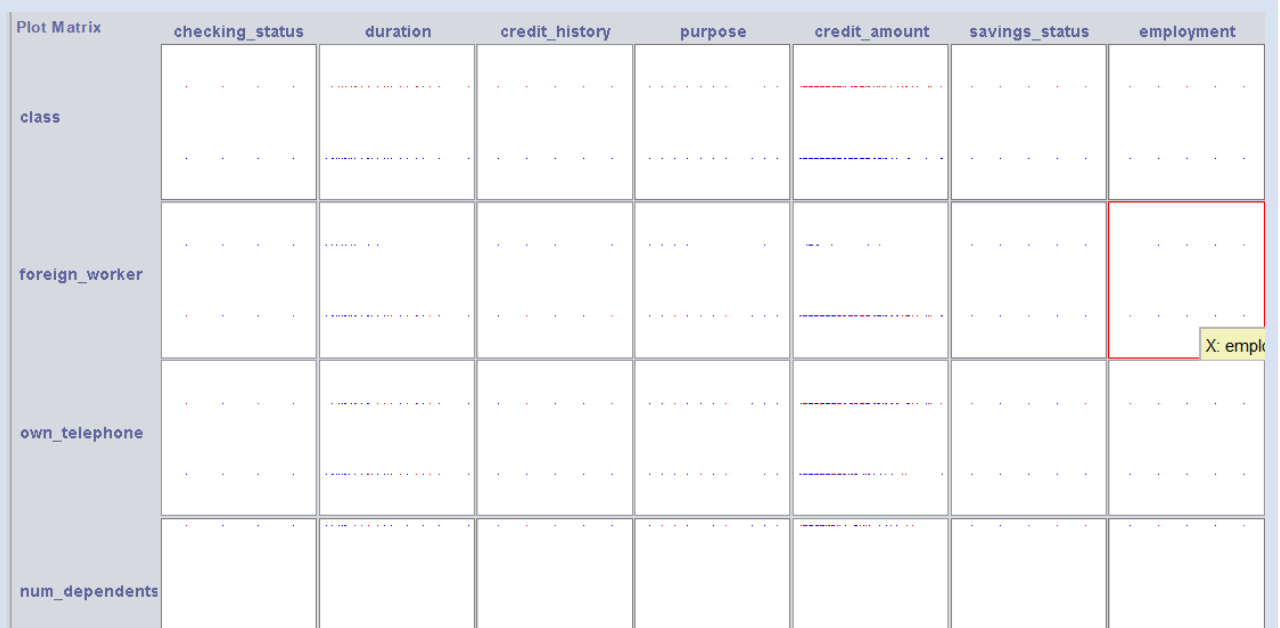


Click on a panel and you will see a bigger version of the graph.

Activity

For the credit.arff data use the visualisation tab. Produce a screenshot for plot of age against credit amount. Does there appear to be a relationship between these two features?

Answer



Once you have got a feeling for the data, move on to using machine learning tools to learn and classify the data.

Task c)



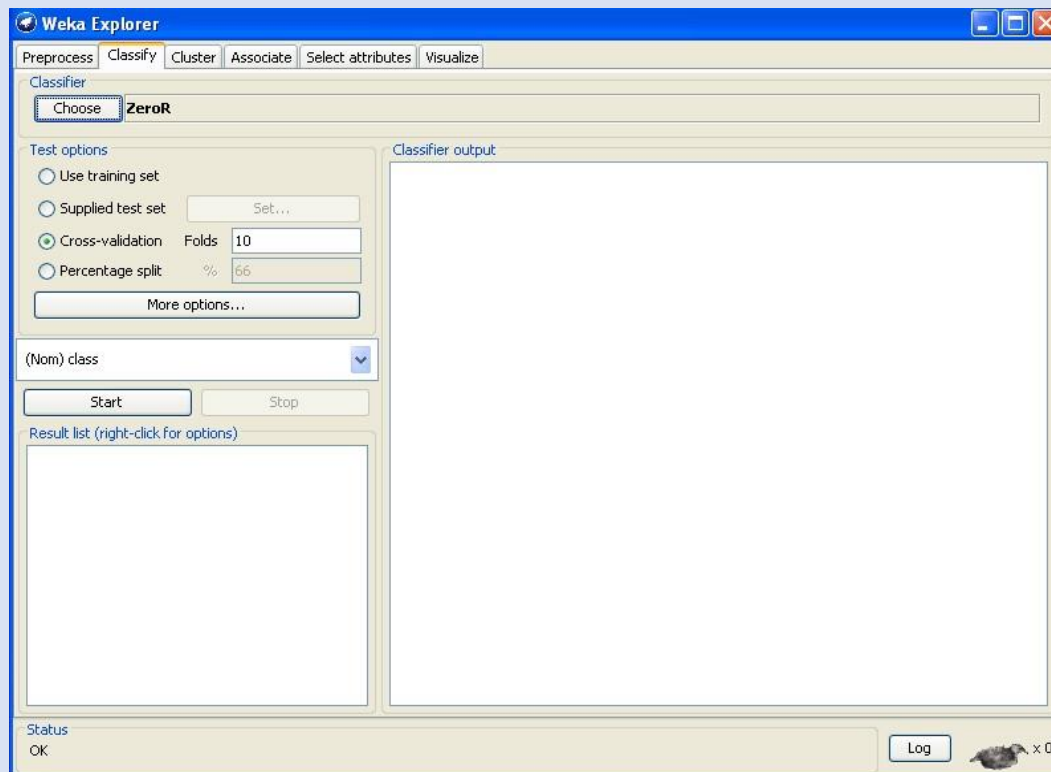
WEKA Intelligent data mining techniques

We will now look at how we can get the machine learning approaches to learn data and recognise samples it as not seen before.

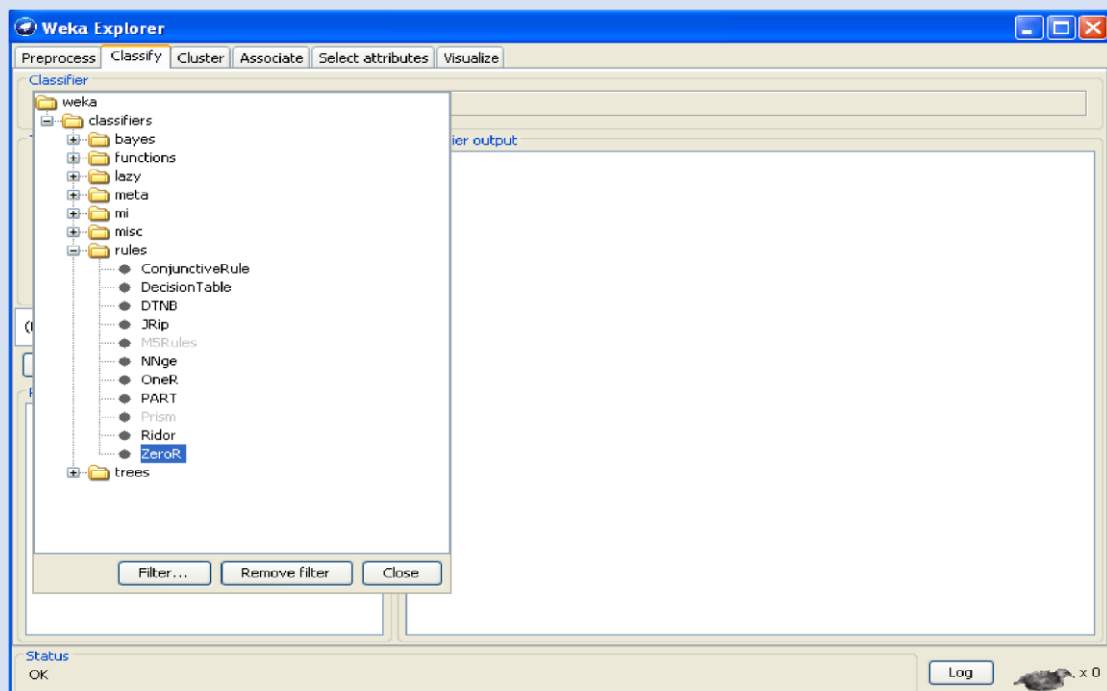
Ensure you have opened the iris data iris.arff.

Select the classify tab

Click the choose button



There are numerous classifiers to choose from. Each is backed by a specific way of classifying the data. The best approach is very much dependant on the data you are using.



Task d)



Decision Tree

Select trees and J48

This is a decision tree algorithm.

Select under test option cross-validation 10 folds split.

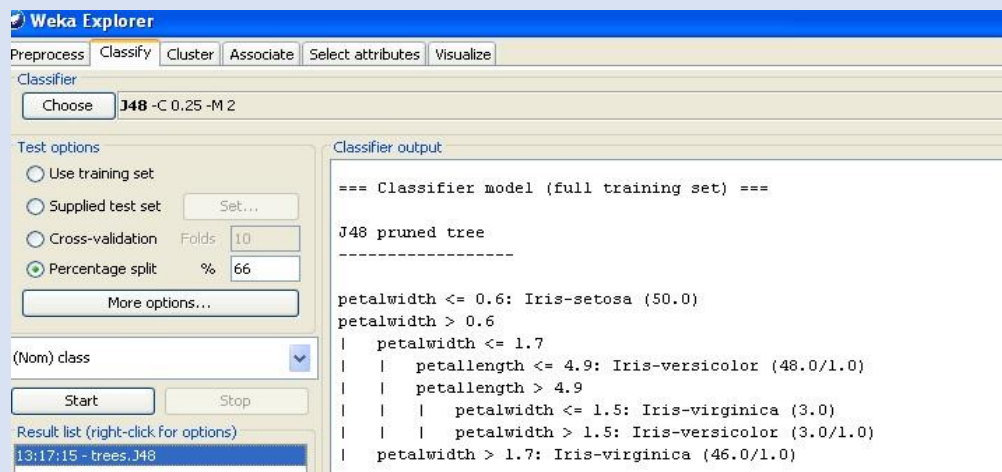
The 10 folds split approach takes the data split it into 10 chunks and uses 9 chunks to train the network and the 10th chunk to test it. It repeats this 10 times with a different chunk for testing each time.

Click start

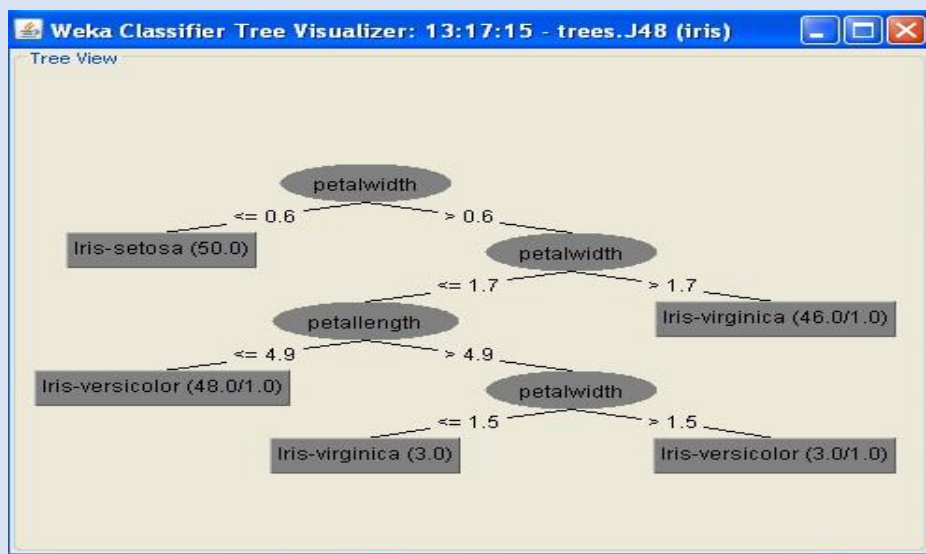
The results appear in the table on screen.

How well did the tree algorithm do in classifying iris?

Right click on the result list and select visualize tree



A tree is created that shows the rules that it used to classify the iris into different types.



Task f)



Neural Networks

Data mining would not be complete without neural networks. In this example we will use the multilayer perceptron neural network (MLP) to classify the iris data.

Do some research on the web on what a neural network and is and what a multilayer perceptron is. Keep it simple!!

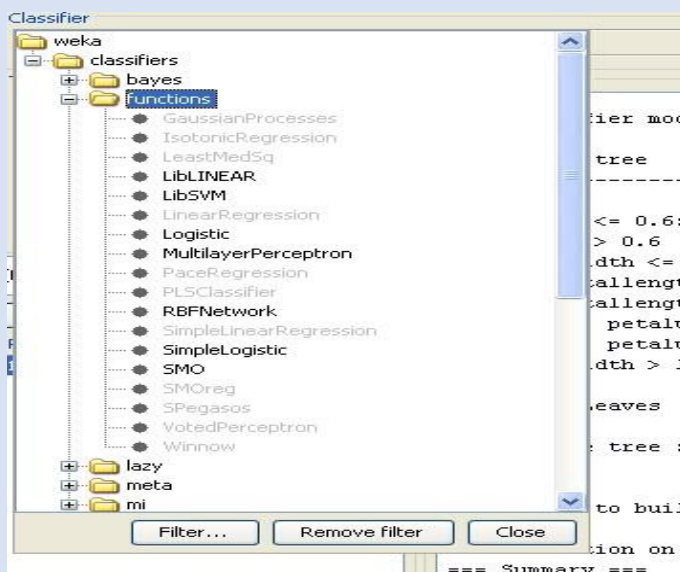
Ensure you have opened the iris data iris.arff.

Select the classify tab

Click the choose button

Select Functions

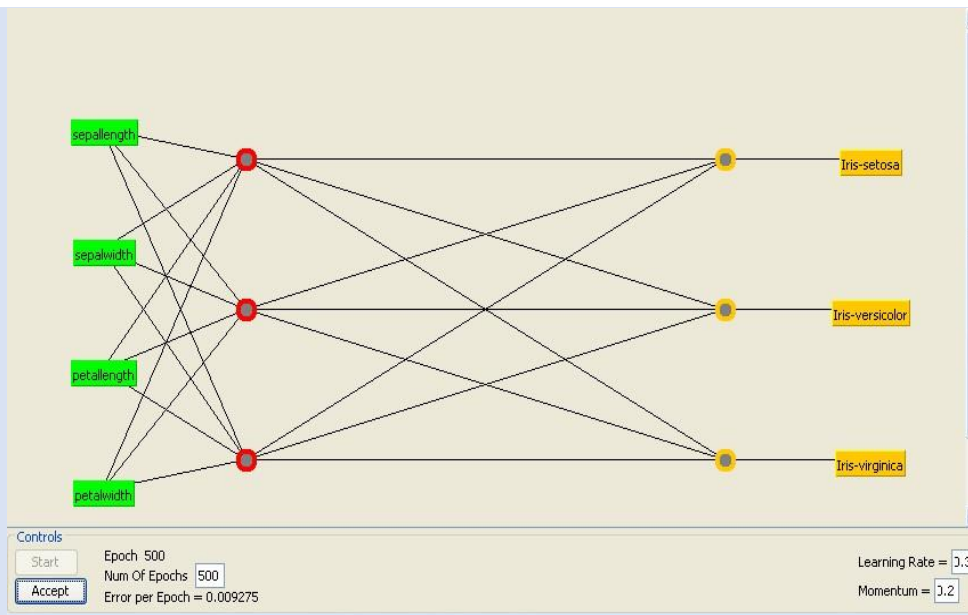
Select MultilayerPerceptron



In the choose line you will see the various parameters used by the multilayer perceptron such as training time.

Right click on the choose line and select show properties and you can see each of these parameters.

Just press OK at the moment



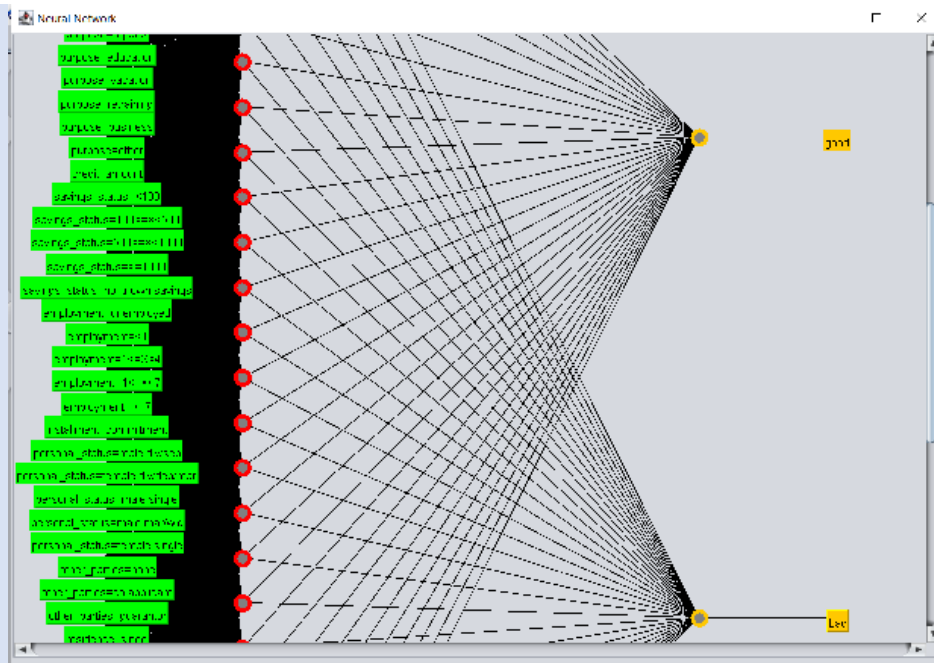
Activity

With the credit data create a MLP neural network (leave all parameters unchanged).

Create a screenshot of the network's performance. Describe in a couple of sentences how successful the MLP is in classifying unseen samples of people's credit worthiness. Look at the confusion matrix to see how many it gets wrong or right.

Currently in the hidden layer the value is a, which sets the number of units in the hidden layer to the $(\text{number of attributes} + \text{number of classes}) / 2$. Try using the value 8 instead. Also reduce the training time to 200. Press start and create a screen shot showing the performance of this network. Was it better or worse than before? How does changing these parameters influence the performance of the neural network?

Answer



=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.809	0.483	0.796	0.809	0.802	0.329	0.734	0.841	good
	0.517	0.191	0.536	0.517	0.526	0.329	0.734	0.573	bad
Weighted Avg.	0.721	0.396	0.718	0.721	0.719	0.329	0.734	0.760	

=== Confusion Matrix ===

```

a  b  <-- classified as
566 134 |  a = good
145 155 |  b = bad

```

8 Hidden Units and 200 epochs

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.794	0.520	0.781	0.794	0.788	0.278	0.717	0.846	good
	0.480	0.206	0.500	0.480	0.490	0.278	0.717	0.516	bad
Weighted Avg.	0.700	0.426	0.697	0.700	0.698	0.278	0.717	0.747	

=== Confusion Matrix ===

```

a  b  <-- classified as
556 144 |  a = good
156 144 |  b = bad

```

Task f)



Clustering

Clustering is the process by which the machine learning approach positions like data in the same location on a map. The advantage of clustering approaches is they do not need to be told the correct classes they work it out themselves.

Ensure the iris.arff data is open

Select the cluster tab

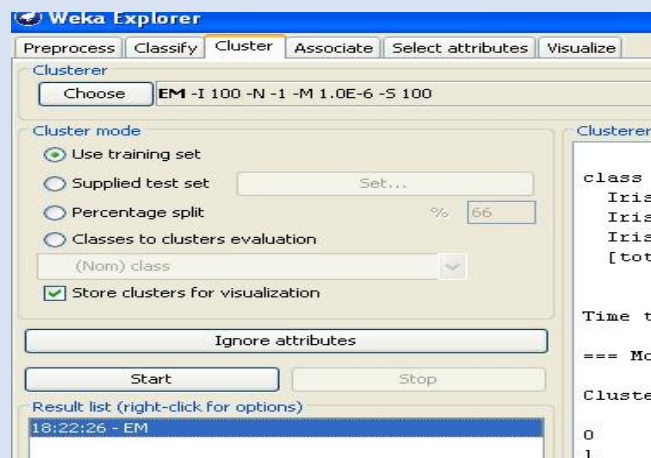
Select the EM clustering technique

Click start

Activity

How many clusters does it create? Do they match the different iris?

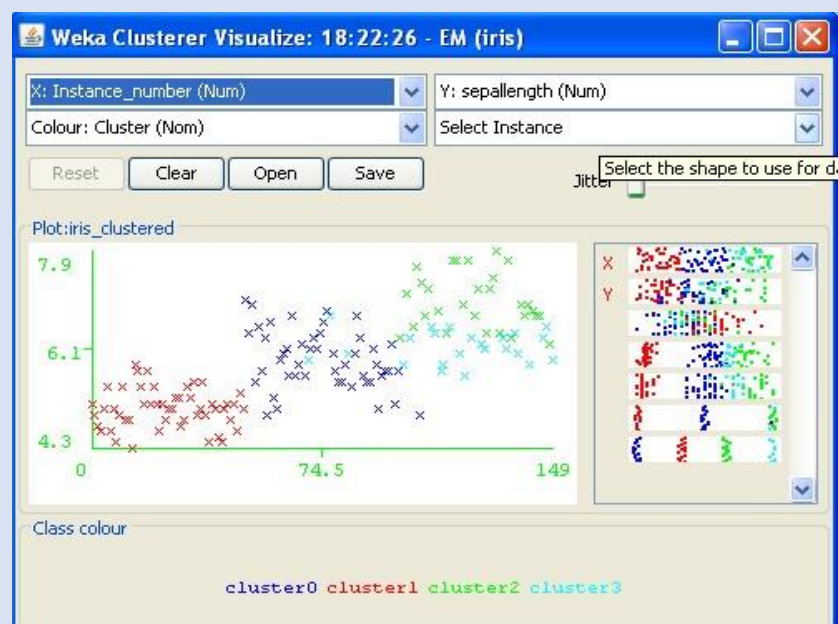
Under results list right click on the EM



Select visualize clusters assigned

It shows the clusters that have been created for the iris data.

If you double click on one of the crosses it shows you what the original data point was.



Activity

Try this clustering approach with the credit.arff data.

Create a screenshot showing the visualisation of the clusters. Does the clustering method create two clear groups that match whether a person is or is not creditworthy?

